LESSON 07 · CHEAT SHEET

Modules, venv & pip

VIRTUAL ENVIRONMENT (ONCE PER PROJECT)

```
python -m venv .venv          # create
.venv\Scripts\activate        # prompt shows (.venv)
deactivate                    # leave it
```

New terminal = activate again. "Scripts is disabled"? `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once.

PIP (WITH THE VENV ACTIVE)

```
pip install rich              # one package
pip install -r requirements.txt   # the list
pip list                      # what's installed
pip show rich                 # version, where
pip freeze > requirements.txt # save the list
```

Package names in `requirements.txt`, one per line. `.venv/` never goes into git.

THREE WAYS TO IMPORT

```
import math
math.ceil(2.1)                3

from datetime import date, timedelta
date(2026, 4, 5)              # no prefix

import statistics as st       # nickname
st.mean([1, 2, 3])            2
```

Imports go at the top of the file. `help(math)` in the REPL is the manual; `q` leaves it.

YOUR OWN MODULE

```
# my_work/paytools.py
"""One-line description."""
def monthly(annual):
    """Docstring: what it does."""
    return round(annual / 12, 2)

if __name__ == "__main__":
    print("self-test")        # only when run directly

# my_work/report.py  (same folder!)
import paytools
paytools.monthly(32500)       2708.33
```

MATH & STATISTICS

`math.ceil(2.1)` / `floor`	3 / 2 (ROUNDUP / INT)
`math.sqrt(16)`	4.0
`math.pi`	3.14159...
`statistics.mean(xs)`	AVERAGE
`statistics.median(xs)`	MEDIAN
`statistics.stdev(xs)`	STDEV.S
`statistics.mode(xs)`	MODE

DATETIME

```
from datetime import date, timedelta
d = date(2026, 4, 5)
date.fromisoformat("2017-03-13")
date.today()                  # real today
(d - start).days              3310   # int
d + timedelta(days=90)
d.year, d.month, d.day
f"{d:%d %b %Y}"               05 Apr 2026
f"{d:%A}"                     Sunday
d.strftime("%d/%m/%Y")        05/04/2026
```

`%d %m %Y`	05 04 2026	`%b %B`	Apr April
`%a %A`	Sun Sunday	`%H:%M`	14:30

RANDOM

```
import random
random.seed(1019)             # repeatable
random.choice(names)          # one
random.sample(names, 3)       # three, no repeats
random.shuffle(names)         # in place
random.randint(1, 6)          # 1..6 inclusive
```

GOTCHAS

- "No module named X": venv not active, or not installed in it
- Never name a file `random.py`, `csv.py`, `math.py`: it hides the real one
- `import Math` fails: module names are case-sensitive
- `today - start` is a `timedelta`; you want `.days`
- Import a module from the same folder as the file you run